SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA0302	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
Student Name:	Nissanki Sri Venkata Lakshmi Kavya	**Reg. No.:**	192373036
Section / Batch:	Slot D	**Date:**	30-07-2026

Code of Conduct

I Nissanki Sri Venkata Lakshmi Kavya (192373036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:N.S.V.L.Kavya

Instructions

- This test contains ONE question.
- Duration: 30 minutes.

Annexure A - QUESTIONS

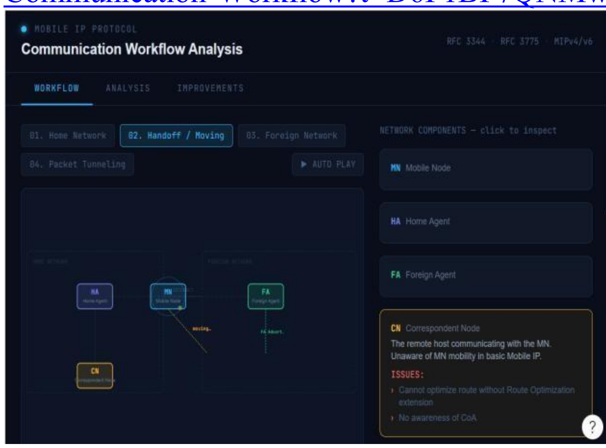
1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

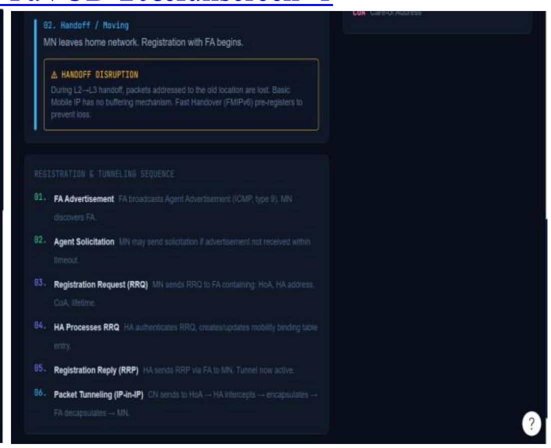
Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).
- Figma Link: https://www.figma.com/make/RceqUPSWSkqVGxaqJ4ebUw/Mobile-IP-Communication-Workflow?t=D6F1BF7QNMwYaVOD-20&fullscreen=1
- 



2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

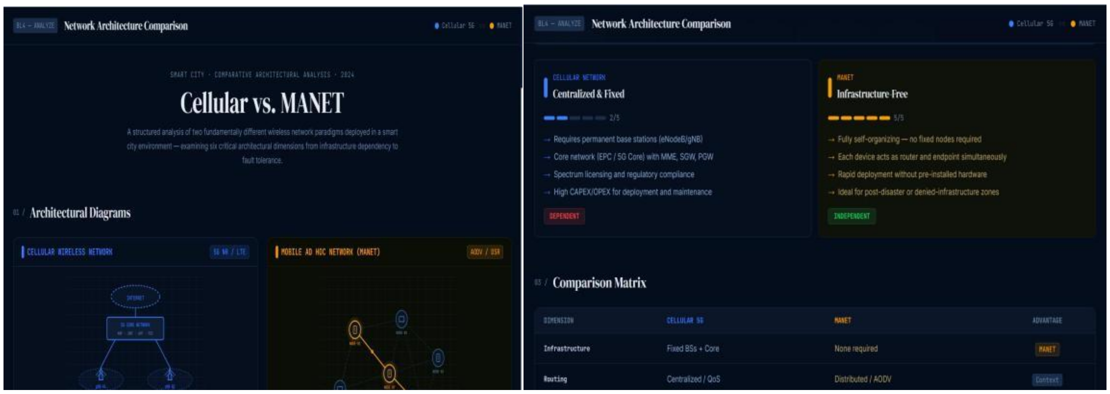
Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.
- https://www.figma.com/make/oWMgAGi7svhr71Vf45N74z/Comparative-Network-Architecture-Diagram?t=NBINiLOnhmjHC3WZ-20&fullscreen=1
- 

3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

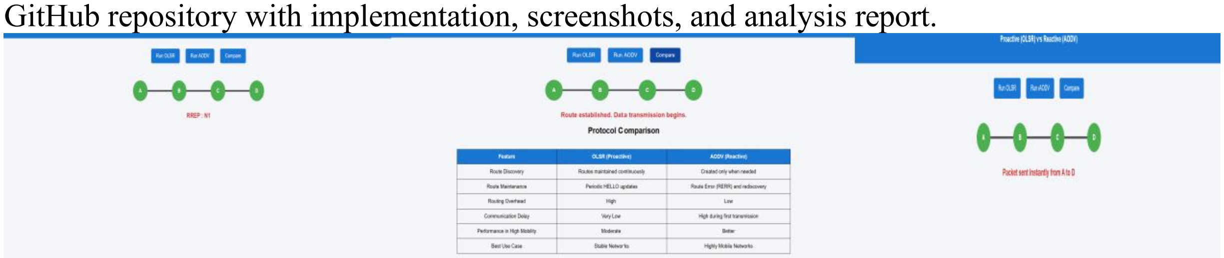
Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Deliverables:

- Source code/design files.
- GitHub repository with implementation, screenshots, and analysis report.
- 

Feature	OLSR (Proactive)	AODV (Reactive)
Route Discovery	Routes maintained continuously	Created only when needed
Route Maintenance	Periodic HELLO updates	Route Error (RERR) and rediscovery
Routing Overhead	High	Low
Communication Delay	Very Low	High during first transmission
Performance in High Mobility	Moderate	Better
Best Use Case	Stable Networks	Highly Mobile Networks

4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

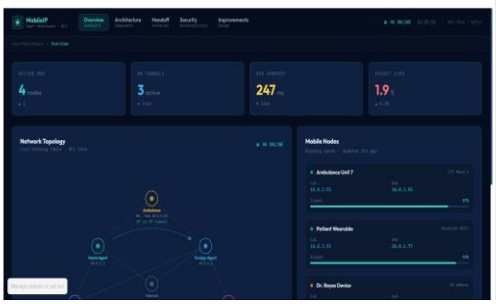
Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.

Analyze:

- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.
- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.

Deliverables:

- Interactive prototype.
- GitHub repository containing design files, exported prototype, and documentation.
- Figma Link: https://www.figma.com/make/RyY8vsw8rjaukzFBsP7mo5/Smart-Healthcare-Mobile-IP-Design?t=D6F1BF7QNMwYaVOD-20&fullscreen=1
- 

5. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

The dashboard should analyze multiple parameters such as:

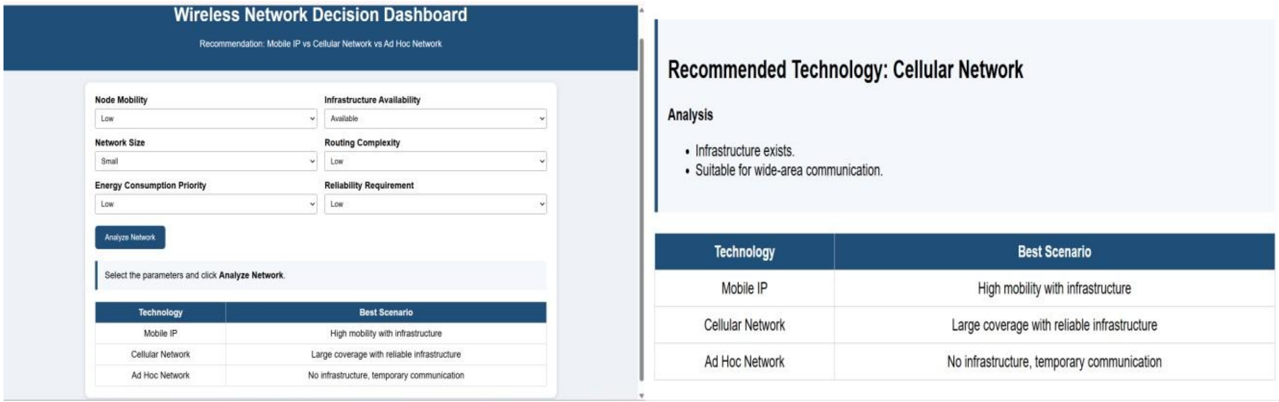
- Node mobility.
- Infrastructure availability.
- Network size.
- Routing complexity.
- Energy consumption.
- Reliability requirements.

The system should justify its recommendation based on the analyzed parameters.

Deliverables:

- Source code or Figma project.

- GitHub repository with complete code/design, README, screenshots, and analysis report.



Criterion	**Marks**
Design Quality (Figma/Programming Tool)	20
Technical Analysis (BL4)	30
Correctness of Mobile Networking Concepts	20
GitHub Repository Organization & Documentation	15